

Lab - 12 : The Solow model of economic growth (Robert Solow)

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 CS302 - Modelling and Simulation*

I. Equations

$$Y(t) = F(C, L) = C^\alpha L^{1-\alpha}, \quad 0 < \alpha < 1$$

Labor Growth (Harrod's Natural Growth Model)

Labor grows exponentially:

$$\dot{L} = nL, \quad L(t) = L_0 e^{nt}, \quad 0 < n < 1$$

Capital Growth

The rate of capital accumulation is proportional to the savings from production:

$$\dot{C} = sY, \quad 0 < s < 1$$

Capital-to-Labor Ratio Dynamics

By defining $r = \frac{C}{L}$, the dynamics of r are described by:

$$\dot{r} = sr^\alpha - nr$$

Integral Solution for $r(t)$

The analytical solution to the above differential equation is:

$$r(t) = \left(\frac{s}{n}\right)^{\frac{1}{1-\alpha}} \left[1 + Ae^{-n(1-\alpha)t}\right]^{\frac{1}{1-\alpha}}$$

where A is an integration constant given by:

$$A = \left(\frac{n}{s}\right) r_0^{1-\alpha} - 1$$

Transition Time

The transition from early power-law growth to stabilization occurs at:

$$t_{\text{trans}} = \frac{1}{n(1-\alpha)} \ln\left(\frac{-A}{1-\alpha}\right)$$

II. Graphs

The following parameter values were used throughout the assignment:

- $s = 0.15$ (savings rate)
- $n = 0.03$ (labour growth rate)
- $\alpha = 0.5$ (Cobb-Douglas parameter)

A. Question 1

We plot $\dot{r}$ versus r to analyze the phase diagram, identify fixed points, and verify intersections by graphing $y_1 = sr^\alpha$ and $y_2 = nr$.

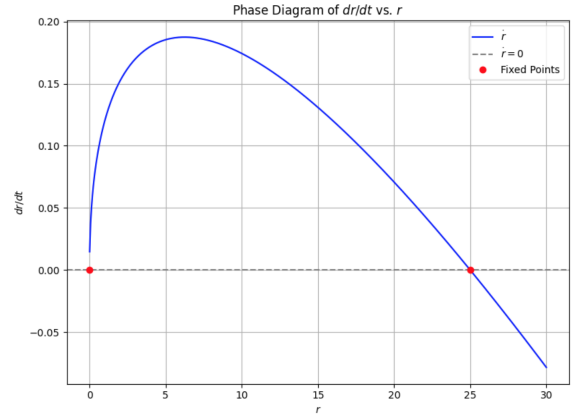


Fig. 1 Phase Diagram of $\dot{r}$ vs r

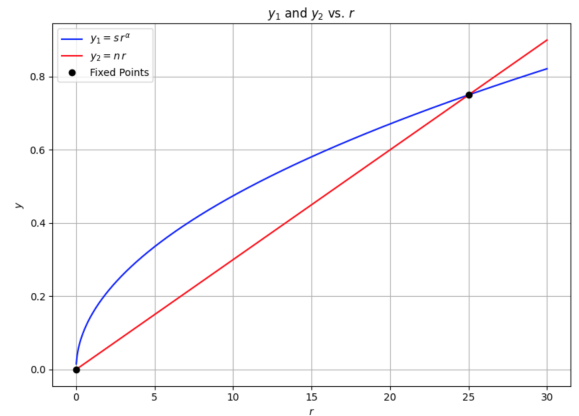


Fig. 2 Graphs of y_1 and y_2 vs r

The phase diagram reveals two fixed points, which correspond to the intersections of y_1 and y_2 , aligning with theoretical predictions.

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B. Question 2

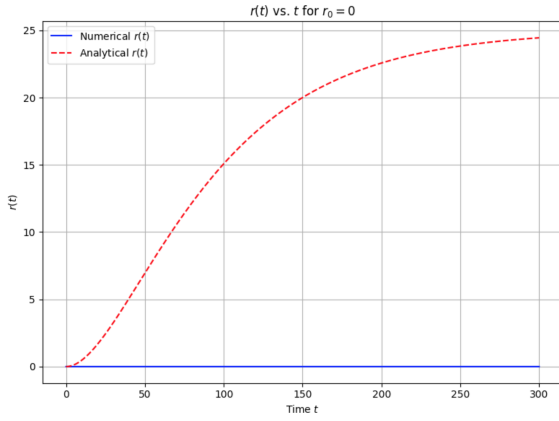


Fig. 3 Numerical and Analytical Solutions with $r_0 = 0$ and $\Delta t = 0.033$. $t_{trans} = 46.21$ s (normal scale)

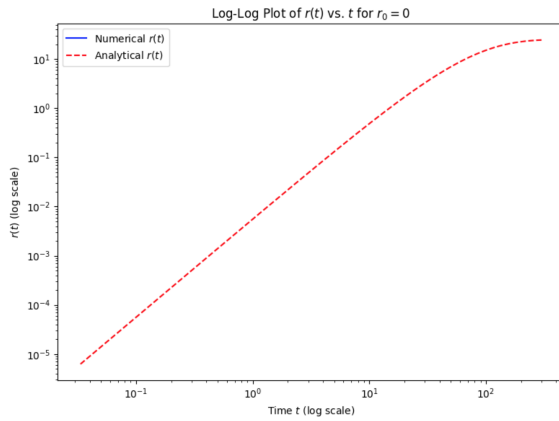


Fig. 4 Comparison of Numerical and Analytical Solutions for $r(t)$ with $r_0 = 0$ and $\Delta t = 0.033$. Transition time $t_{trans} = 46.21$ s (log-log scale)

When $r(0) = 0$, One limitation of numerical integration is that Euler's approach stops at zero while the analytical answer grows as predicted.

C. Question 3

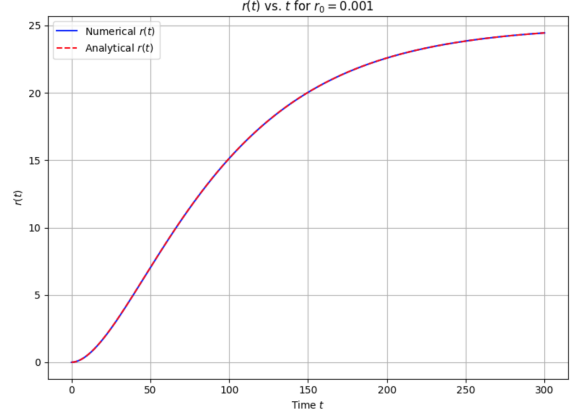


Fig. 5 Comparison of Numerical and Analytical Solutions for $r(t)$ with $r_0 = 0.001$ and $\Delta t = 0.033$. Transition time $t_{trans} = 45.79$ (normal scale)

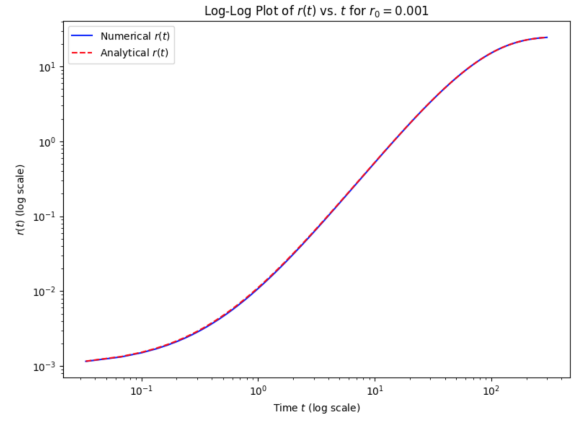


Fig. 6 Comparison of Numerical and Analytical Solutions for $r(t)$ with $r_0 = 0.001$ and $\Delta t = 0.033$. Transition time $t_{trans} = 45.79$ (log-log scale)

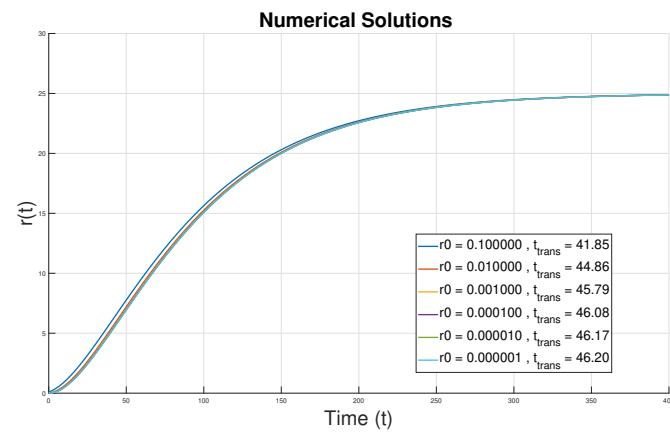


Fig. 7 Comparison of Numerical Solutions for different values of $r(0)$ (normal scale)

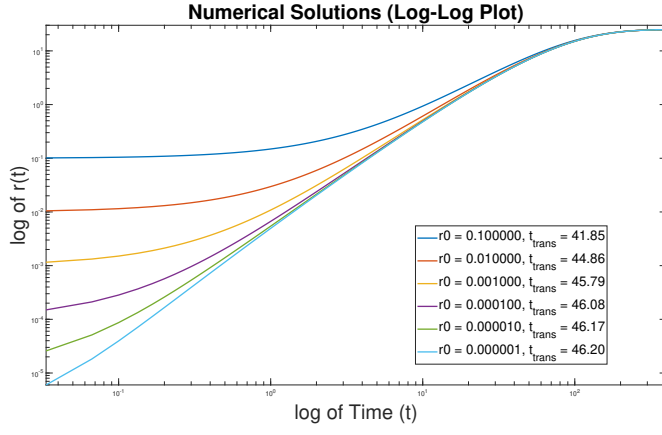


Fig. 8 Comparison of Numerical Solutions for different values of $r(0)$ (log-log scale)

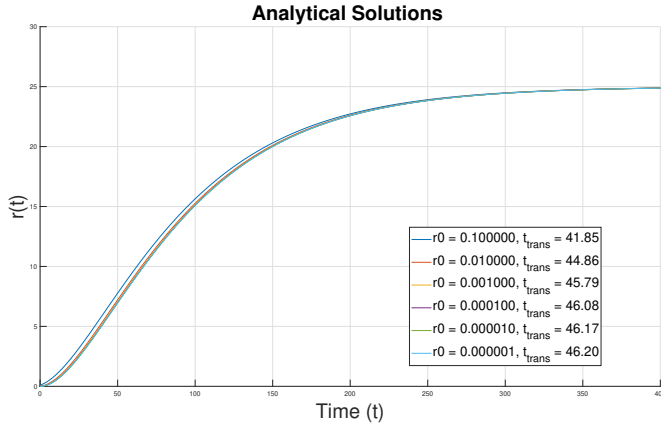


Fig. 9 Comparison of Analytical Solutions for different values of $r(0)$ (normal scale)

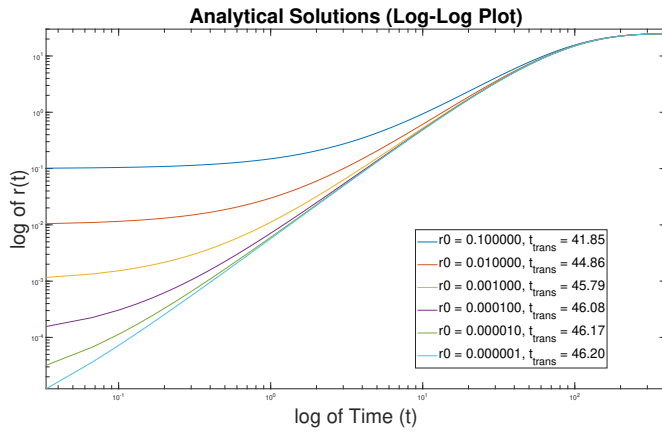


Fig. 10 Comparison of Analytical Solutions for different

values of $r(0)$ (log-log scale)

III. Conclusions

- Initial conditions significantly influence the Solow growth model's performance. When $r(0) = 0$, early growth dynamics are poorly captured due to limitations in Euler's numerical method.
- For case where $r(0) \neq 0$, the model demonstrates consistency, as numerical and analytical results closely align, affirming its accuracy in realistic scenarios.
- The steady-state convergence of both approaches confirms the model's robustness and reliability in forecasting long-term growth behavior.